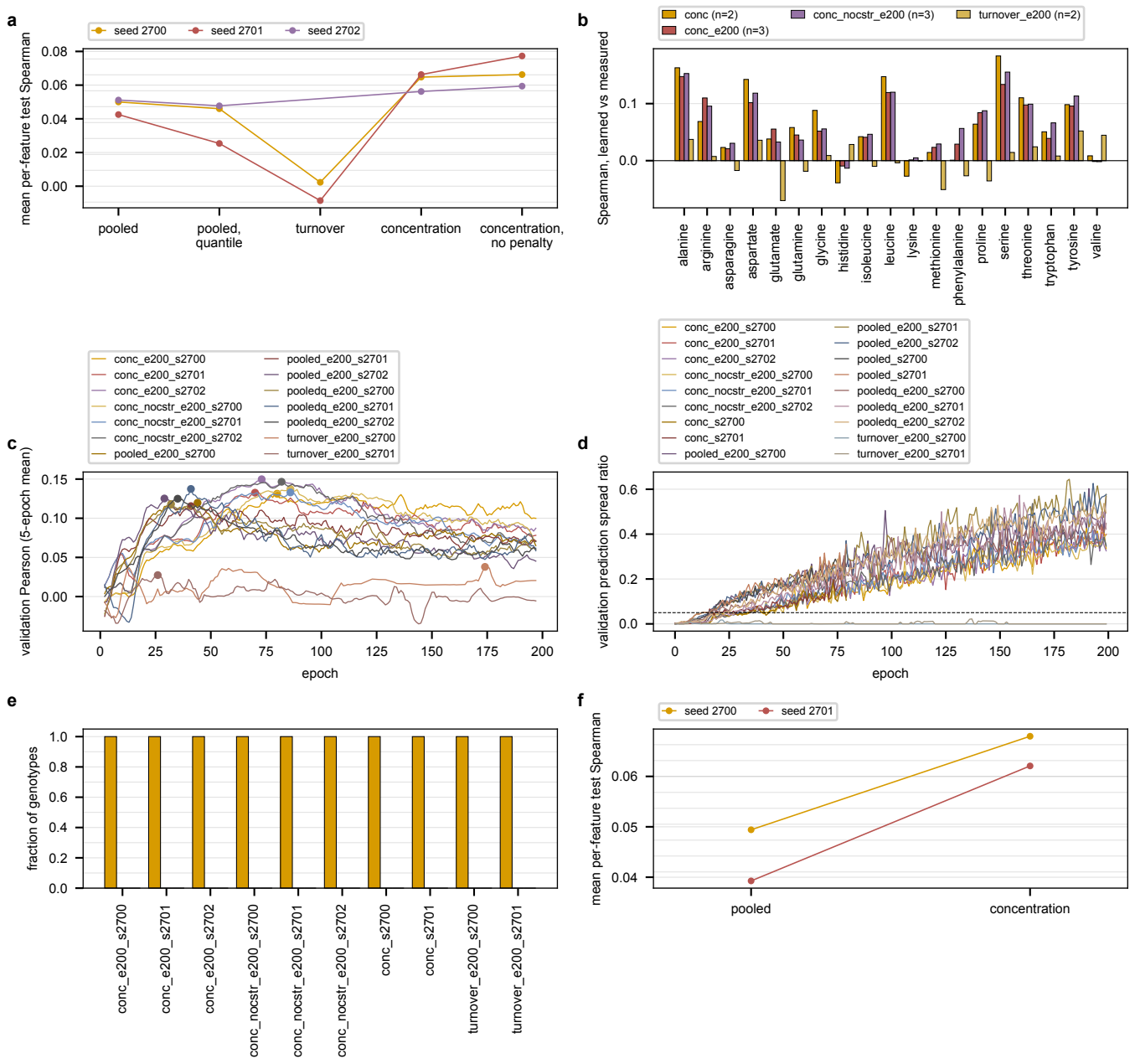

a
seed 2700
seed 2701
seed 2702
mean per-feature test Spearman
0.08
0.06
0.04
0.02
0.00
pooled
pooled, quantile
turnover
concentration
concentration, no penalty
b
conc (n=2)
conc_e200 (n=3)
conc_nocstr_e200 (n=3)
turnover_e200 (n=2)
Spearman, learned vs measured
0.1
0.0
alanine
arginine
asparagine
aspartate
glutamate
glutamine
glycine
histidine
isoleucine
leucine
lysine
methionine
phenylalanine
proline
serine
threonine
tryptophan
tyrosine
valine
c
conc_e200_s2700
conc_e200_s2701
conc_e200_s2702
conc_nocstr_e200_s2700
conc_nocstr_e200_s2701
conc_nocstr_e200_s2702
pooled_e200_s2700
pooled_e200_s2701
pooled_e200_s2702
pooledq_e200_s2700
pooledq_e200_s2701
pooledq_e200_s2702
turnover_e200_s2700
turnover_e200_s2701
validation Pearson (5-epoch mean)
0.15
0.10
0.05
0.00
0
25
50
75
100
125
150
175
200
epoch
d
conc_e200_s2700
conc_e200_s2701
conc_e200_s2702
conc_nocstr_e200_s2700
conc_nocstr_e200_s2701
conc_nocstr_e200_s2702
conc_s2700
conc_s2701
pooled_e200_s2700
pooled_e200_s2701
pooled_e200_s2702
pooled_s2700
pooled_s2701
pooledq_e200_s2700
pooledq_e200_s2701
pooledq_e200_s2702
turnover_e200_s2700
turnover_e200_s2701
validation prediction spread ratio
0.6
0.4
0.2
0.0
0
25
50
75
100
125
150
175
200
epoch
e
fraction of genotypes
1.0
0.8
0.6
0.4
0.2
0.0
conc_e200_s2700
conc_e200_s2701
conc_e200_s2702
conc_nocstr_e200_s2700
conc_nocstr_e200_s2701
conc_nocstr_e200_s2702
conc_s2700
conc_s2701
turnover_e200_s2700
turnover_e200_s2701
f
seed 2700
seed 2701
mean per-feature test Spearman
0.06
0.05
0.04
pooled
concentration